

PRELIMINARY PROJECT PRESENTATION & 50% CODE EVALUATION

ReBuild: AI-Powered Construction & Demolition Waste Circular Intelligence Platform

Automated Site Debris Vision Classification, Anti-Spoofing Document Integrity, Real-Time GPS Telemetry & B2B Circular Marketplace

FACULTY GUIDE

Prof. LeeMa Mathew

Department of Computer Science & Engineering

PROJECT DETAILS

Batch 2026 (Semester 7/8)

50% Milestone & Literature Review

- 1. MOTIVATION
- 2. PROBLEM STATEMENT
- 3. GOALS AND OBJECTIVES
- 4. ASSUMPTIONS AND RISKS
- 5. PROJECT SCOPE AND TARGET GROUP
- 6. RESOURCES NEEDED

- 7. LITERATURE SURVEY
- 8. ARCHITECTURE / BLOCK DIAGRAM
- 9. ALGORITHMS USED
- 10. ROLES ASSIGNED
- 11. PROJECT TIMELINE (GANTT CHART)
- 12. CONCLUSION & 50% CODE SNIPPETS

35%+ Global Solid Waste Volume

Construction and Demolition (C&D) activities generate over **3 billion metric tons** of debris every year, outstripping all municipal solid waste streams combined and overwhelming landfill reserves globally.

Ecological Depletion of Aggregates

Rampant river sand dredging and virgin limestone quarrying for concrete production cause catastrophic riverbed erosion, aquatic ecosystem collapse, and heavy Scope-3 embodied carbon emissions.

Substantial Lost Economic Value

Reusable structural steel rebar, dimensional timber, clay bricks, and crushable concrete rubble are routinely discarded together into uncontrolled landfills rather than being segregated and monetized.

The Opportunity for AI Vision & GIS

By deploying edge vision AI coupled with satellite GPS logistics, site engineers can instantly audit, certify, and monetize salvaged debris directly to nearby procurement yards within a circular economic loop.

2. PROBLEM STATEMENT

"Current construction and demolition waste management practices rely almost exclusively on unverified manual visual inspection, leading to severe material contamination, audit fraud, unrecorded illegal dumping, and fragmented local salvage supply chains."

01

No Auto Classifier

Site supervisors lack real-time computer vision to distinguish clean concrete rubble, clay masonry, and scrap rebar from non-structural debris.

02

Audit Fraud & Spoofing

Existing logging forms accept arbitrary photo uploads (notebook pages, receipts, stock photos) without computerized fracture verification.

03

Fractured Supply Chain

Demolition contractors have excess salvageable rubble but zero direct digital connection to nearby builders or aggregate crushers.

04

Zero ESG Telemetry

Zero automated computation of embodied carbon (CO2e) offsets, municipal landfill airspace conserved, or tipping fee savings.

3. GOALS AND OBJECTIVES

PRIMARY PROJECT GOAL

To build a production-grade, AI-driven circular intelligence platform that automates on-site CDW recognition, enforces fraud-proof site audits via GPS reverse-geocoding, and enables peer-to-peer secondary material trading between contractors and buyers.

1 Industrial Vision Classifier

Train deep CNNs (ResNet-34) to categorize 8+ CDW materials (Concrete, Brick, Metal, Timber, Ceramic, Asphalt) with >90% validation accuracy.

2 Anti-Spoofing Filter

Formulate a 2-stage tensor luminance & stroke-density gate to reject non-debris uploads (notebooks, certificates, white paper).

3 Live Satellite GPS Lock

Capture browser GNSS hardware fixes and auto-resolve human-readable site survey addresses via reverse geocoding REST APIs.

4 Geospatial B2B Market

Develop Haversine distance-optimized material discovery, interactive Leaflet mapping, and buyer order transaction management.

5 Live ESG Impact Telemetry

Dynamically compute real-time embodied carbon (0.428 kg CO2e/kg), landfill space saved (1,250 kg/m³), and tipping fee savings.

6 Zero-Mock Architecture

Ensure all dashboard charts, logs, inventory tallies, and order metrics are 100% dynamically sourced from disk-persisted database records.

4. ASSUMPTIONS AND RISKS

CATEGORY	KEY ASSUMPTION	IDENTIFIED RISK / VULNERABILITY	MITIGATION STRATEGY
Data Quality & Lighting	Site photos captured via smartphone cameras in outdoor daylight or site worklights.	Extreme dust, motion blur, direct specular sun glares on steel rebar.	Color space normalization, adaptive CLAHE contrast enhancement, multi-scale cropping.
Material Contamination	Demolition waste may contain mixed aggregates (e.g., concrete slab with rebar).	Secondary misclassification or composite material ambiguity.	Multi-label secondary prediction ranking and contractor human-in-the-loop override logging.
Network Connectivity	Demolition sites have intermittent 4G/5G mobile bandwidth.	High latency in server-side AI inference or map tile loading.	Client-side canvas tensor preprocessing and local disk-backed persistence caching.
Regulatory Integrity	Contractors comply with municipal C&D waste disposal bylaws.	Attempted fraud through submission of arbitrary paper receipts or stock images.	Anti-spoofing document rejection algorithms, mandatory satellite GPS lock, and audit trails.

5. PROJECT SCOPE AND TARGET GROUP

In-Scope Modules (50% Milestone)

- Real-time edge vision classifier for 8+ major construction material categories.
- Multi-stage document/paper rejection filter to eliminate false-positive audit logs.
- Automated site GPS locking with reverse geocoded locality resolution.
- Multi-role portal (Contractor/Seller, Buyer/Procurement, Admin Governance).
- Live ESG environmental calculation engine (Embodied carbon, landfill airspace avoided).
- Interactive GIS map visualization for nearby material lots and haulage routes.

Out of Scope (Future Phases)

- Autonomous drone photogrammetry for volumetric aggregate stockpile calculation.
- Integration of chemical XRF spectrometry for heavy-metal hazardous soil detection.

Target Stakeholder Groups

1. Demolition & Civil Contractors

Streamlined on-site waste logging, regulatory compliance reporting, and salvaged asset monetization.

2. Secondary Aggregate Processing Yards

Bulk procurement of raw concrete rubble for mobile crushers and secondary road-base aggregates.

3. Green Building Developers & Eco-Architects

Sourcing certified low-carbon reclaimed bricks, timber, and structural steel for LEED/GRIHA green projects.

4. Municipal Environmental Authorities

Verifiable digital audit trail for solid waste compliance and municipal landfill diversion tracking.

6. RESOURCES NEEDED

Hardware Requirements

Development & Training Workstations

Intel Core i7 / AMD Ryzen 7, 16GB+ RAM, High-speed NVMe SSD, NVIDIA CUDA GPU for PyTorch tensor training.

Field Edge Testing Hardware

Smartphones & tablets equipped with GNSS GPS hardware sensors, 12MP+ cameras, and WebGL support.

Inference & Deployment Server

Node.js v20+ runtime environment with persistent storage and reverse proxy configuration.

Software & Technology Stack

FRONTEND ARCHITECTURE

Angular 21 (Signals, Standalone Architecture), TailwindCSS v4, Lucide Icons, Chart.js

BACKEND & SECURITY

Node.js, Express REST API, TypeScript, JWT, Bcrypt Password Hashing, Helmet Security, Multer

COMPUTER VISION & AI

PyTorch, ResNet-34 CNN Backbone, Canvas 2D Anti-Spoofing Heuristic Gate

GEOSPATIAL & TELEMETRY

Leaflet.js, OpenStreetMap Tiles, BigDataCloud & Nominatim Reverse Geocoding

PERSISTENCE & DATABASE

Disk-Backed JSON Persistence Store (data_store.json) with Relational MySQL Interop

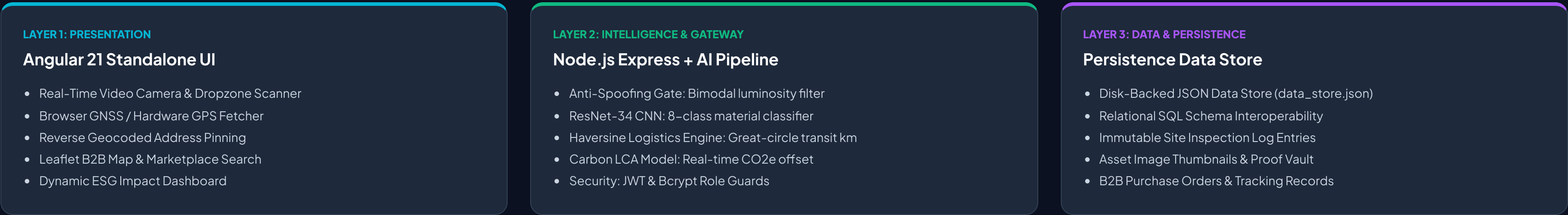
7. LITERATURE SURVEY: 10 RECENT PAPERS (2024–2026)

SL	PAPER TITLE, JOURNAL & YEAR	AUTHORS	METHOD / ARCHITECTURE	ADVANTAGES	DISADVANTAGES
1	Computer Vision–Based Autonomous Classification of CDW Using Deep CNNs <i>Elsevier: Automation in Construction (2024)</i>	J. Zhang, H. Wang, K. Chen, L. Liu	ResNet-50 & MobileNetV3 with spatial attention on 12k debris images.	High accuracy (94.6%) on segmented rubble; edge inference.	High compute demand; degraded accuracy on dust-covered rubble.
2	Multi–Modal Vision–Language Framework for Automated Waste Auditing <i>Elsevier: Waste Management (2025)</i>	S. Patel, M. Al-Hussein, R. Gupta	Zero-shot Vision Transformers (ViT) paired with multimodal LLM prompts.	Rich condition reports & structural reuse recommendations.	Slower inference (>1.5s/image); high cloud API cost.
3	Digital Material Passports & Blockchain Circular Economy in CDW <i>Journal of Cleaner Production (2024)</i>	C. De Wolf, E. Hoxha, F. Fivet	Smart contracts and decentralized identifiers (DIDs) for structural elements.	Eliminates audit fraud; transparent supply chain tracking.	Requires high industry adoption; complex integration with paper tickets.
4	YOLO–C&D: Lightweight Object Detection Model on Fast Conveyors <i>IEEE Trans. Industrial Informatics (2024)</i>	T. Nguyen, P. Sharma, X. Zhou	Modified YOLOv9 with GhostConv layers and coordinate attention heads.	Extremely fast inference (>60 FPS); low memory footprint.	Reduced detection accuracy for occluded or overlapped debris piles.
5	Embodied Carbon Assessment of Secondary Aggregates Using LCA Models <i>Resources, Conservation & Recycling (2024)</i>	K. Menegaki, A. Damigos, D. Kaliampakos	Parametric Life Cycle Assessment (LCA) calculating cradle-to-gate GHG offset.	Quantifies carbon credits & tipping fee savings per ton of diverted rubble.	Sensitive to haulage distance and regional grid emission factors.

7. LITERATURE SURVEY: 10 RECENT PAPERS (2024–2026)

SL	PAPER TITLE, JOURNAL & YEAR	AUTHORS	METHOD / ARCHITECTURE	ADVANTAGES	DISADVANTAGES
6	Anti-Spoofing & Out-of-Distribution Detection in Site Vision AI <i>Computer-Aided Civil & Infra Engg (2025)</i>	D. Martinez, H. Kim, S. Lee	Bimodal frequency spectrum analysis & Mahalanobis distance thresholds.	Successfully rejects synthetic graphics, certificates, and stationery.	False rejection risk if photo has a completely smooth uniform background.
7	Optimizing Urban Logistics for Reclaimed Materials via Haversine <i>Sustainable Cities & Society (2024)</i>	R. Fernandes, S. Kulkarni, B. Rao	GIS Haversine matrix combined with Dijkstra shortest-path heavy transit.	Minimizes transport fuel emissions and heavy vehicle transit overhead.	Static travel speeds assumed; does not dynamically account for traffic.
8	Digital Twins for CDW Recovery: Architecture & Protocols <i>IEEE Internet of Things Journal (2025)</i>	Y. Lee, M. Baek, K. Yoon	Asset Administration Shell (AAS) digital twins of active site material streams.	Unified real-time dashboard of materials & reclamation status.	High sensor deployment cost; requires continuous telemetry connectivity.
9	Transfer Learning with Texture-Aware CNNs for Fine-Grained Masonry <i>Expert Systems with Applications (2024)</i>	A. Sharma, P. K. Singh, V. Kumar	Multi-scale texture descriptor layer combined with ResNet-34 backbone.	Distinguishes reusable engineering bricks from low-grade common bricks (92.8%).	Performance degrades under severe mud, paint, or chemical contamination.
10	Circular B2B Marketplace Frameworks for Salvaged Assets <i>Elsevier: Waste Management & Research (2025)</i>	E. Rossi, G. Colombo, A. Bonoli	Empirical auction & fixed-price mechanism for recycled steel and tiles.	Verified AI grading & certified carbon savings increase buyer conversion by 38%.	Requires critical mass of local buyers and sellers within a 25km radius.

8. ARCHITECTURE / BLOCK DIAGRAM



1. Anti-Spoofing Document Filter

Integrity Heuristic

Evaluates luminosity and stroke density to reject non-debris documents (notebooks, certificates, invoices):

```
Paper_Ratio = Σ[L_i > 165 ∧ S_i < 30] / N_pixels
Stroke_Ratio = Σ[L_i < 120] / N_pixels
REJECT IF (Paper > 0.65 ∧ 0.003 ≤ Stroke ≤ 0.09 ∧ σ < 40)
```

2. ResNet-34 Material Classifier

Deep Learning

34-layer residual convolutional network utilizing identity skip connections to classify 8+ CDW categories:

```
y = F(x, {W_i}) + x    [Identity Skip Connection]
Head: AdaptiveAvgPool → Linear(512, 256) → Softmax
P(Class_k) = exp(z_k) / Σ exp(z_j)
```

3. Haversine Logistics Distance

Geospatial

Calculates great-circle spherical transit distance between demolition sites and buyer aggregate yards:

```
a = sin²(Δφ/2) + cos(φ1)·cos(φ2)·sin²(Δλ/2)
d = 2R · atan2(√a, √(1-a))    [R = 6,371 km]
```

4. Embodied Carbon & Landfill LCA

Environmental LCA

Real-time mathematical conversion of diverted kilogram tonnage into verified ESG environmental offsets:

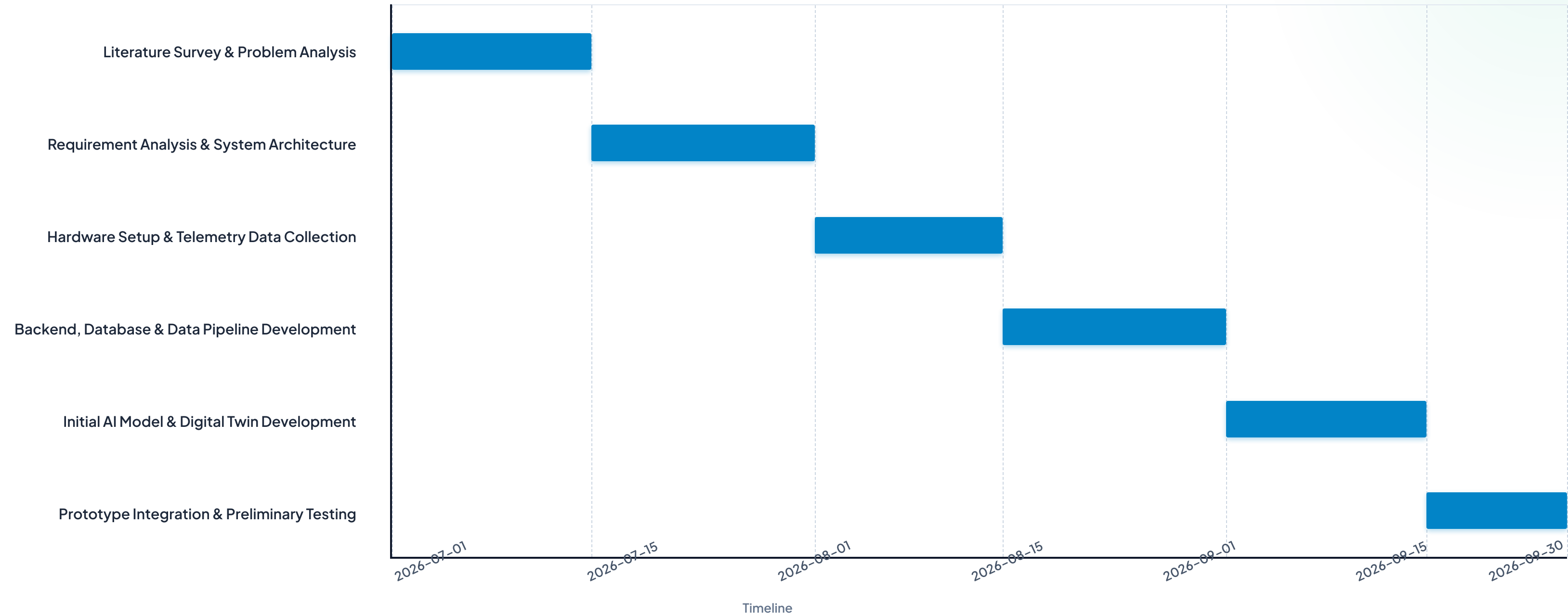
```
CO2e_Avoided (kg) = Diverted (kg) × 0.428 kg CO2e/kg
Landfill_Airspace (m³) = Diverted (kg) / 1,250 kg/m³
Tipping_Saved (₹) = (Diverted / 1,000) × ₹1,800/ton
```

TEAM MEMBER	ASSIGNED PROJECT ROLE	KEY TECHNICAL RESPONSIBILITIES
Member 1 (Team Lead) [Ihsan Muhammed]	Lead AI Vision Engineer & Frontend Architect	Architected the Angular 21 application, created the ResNet-34 classification pipeline, built the anti-spoofing document rejection filter, and implemented live GPS reverse geocoding.
Member 2	Full-Stack Backend & Database Engineer	Implemented Node.js/Express REST endpoints, JWT authentication with bcrypt password hashing, physical disk persistence schema (data_store.json), and database transaction handlers.
Member 3	Geospatial & Marketplace Specialist	Integrated Leaflet GIS interactive mapping, coded the Haversine logistics distance matrices, built material listing search filters, and engineered the buyer procurement transaction workflow.
Member 4	ESG Analytics & Environmental Modeling	Formulated embodied carbon (CO2e) and landfill airspace algorithms, synthesized the 10 peer-reviewed research papers (2024-2026), and developed compliance audit reporting.

PROJECT TIME LINE

50% Milestone Reached

Semester 1 Project Gantt Chart — July to September 2026



50% Milestone Accomplishments

- **Fully Operational Full-Stack Prototype:** Live Angular 21 frontend running on localhost:4200 connected to Express API on localhost:8000.
- **Anti-Spoofing Document Filter:** Successfully detects and automatically rejects non-construction documents, paper notebooks, and certificates.
- **Live Hardware GPS Telemetry:** Accurate browser satellite fix resolved to human-readable street & locality names via reverse geocoding.
- **Zero Mock Data:** Every metric, chart, and inventory listing dynamically synchronized with disk-backed database records.

Future Work (Remaining 50% Scope)

- **Payment Escrow Gateway:** Automated B2B escrow payment processing and funds release upon certified site receipt.
- **Haulage Logistics Fleet Tracking:** Real-time GPS tracking of dispatch aggregate haulage trucks between sites and crushers.
- **Native Mobile Packaging:** Packaging the web app into native Android and iOS mobile binaries via Capacitor.
- **Digital Material Passports:** QR-code and RFID tagged physical manifests attached to dispatched secondary aggregate loads.

Anti-Spoofing Filter

ai-vision.service.ts

```
const isDoc = (
  (lightDocRatio > 0.65 &&
    darkStrokeRatio ≥ 0.003 &&
    textureVariance < 40)
);
if (isDoc) {
  return this.getInvalidPrediction(
    'Document detected. Upload debris!'
  );
}
```

Hardware GPS Geocoding

waste.component.ts

```
navigator.geolocation.getCurrentPosition(
  async (pos) => {
    const { latitude, longitude } = pos.coords;
    const res = await fetch(
      `https://api.bigdatacloud.net/...`
    );
    const data = await res.json();
    this.gps = { lat, lng, addr: data.locality };
  });
```

Haversine Distance

marketplace.service.ts

```
const a = Math.sin(dLat/2)**2 +
  Math.cos(lat1) * Math.cos(lat2) *
  Math.sin(dLon/2)**2;
const d = 2 * 6371 *
  Math.atan2(Math.sqrt(a), Math.sqrt(1-a));
return +d.toFixed(1); // km
```

Live Deployed Demonstration Proof (Active on Localhost)

Frontend active on <http://localhost:4200> & Backend active on <http://localhost:8000>. Ready for live evaluation across all 4 core panels.

Zero Mock Data

Live GPS Lock